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# Automatic Skull Stripping for CT Images of Traumatic Brain Injuries (TBI)

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## Abstract

Traumatic Brain Injury, which is also known as TBI, is a serious global health concern, with neuroimaging playing a vital role in its diagnosis and prognosis. Skull stripping refers to the technique of isolating the brain region by eliminating non-brain structures from neuroimaging data. It is an essential preprocessing step in neuroimaging analysis to improve lesion detection, feature extraction, and tissue classification. This study presents an efficient skull stripping method, ThresContCT\_TBI, specifically designed for computed tomography (CT) images of TBI patients. ThresContCT\_TBI integrates intensity thresholding, morphological processing, contour filtering, and distance transforms to achieve robust brain extraction while addressing challenges such as traumatic lesions and anatomical complexities. The effectiveness of the approach was evaluated using segmentation metrics, including accuracy (98.68%), Dice coefficient (97.79%), intersection over union (IoU) (95.68%), and Hausdorff distance (HD). The results demonstrate the method's effectiveness in preserving brain structures while accurately eliminating non-brain tissues. Compared to deep learning-based and traditional methods, our approach offers computational efficiency and does not require extensive labeled datasets, which makes it ideal for use in real-time applications in healthcare. This research emphasizes the critical role of optimized skull stripping in improving automated neuroimaging workflows, with potential applications in TBI diagnosis and beyond.

**Keywords** Skull stripping · Computed tomography (CT) · Traumatic brain injury (TBI) · Image processing · Deep learning (DL)

## Introduction

TBI is a global health concern caused by external forces or penetrating injuries to the head, affecting neurological function and pathology. Approximately 50 million individuals are affected by TBI each year, and it is responsible for

30–40% of all deaths resulting from injuries globally [1, 2]. Routine neuroimaging immediately after brain injury can provide insights into prognosis, including mortality and morbidity. Ultrasound (US), Computed tomography (CT), and magnetic resonance imaging (MRI) are common neuroimaging techniques used in the clinical evaluation of TBI. These techniques can help detect acute brain injuries such as hemorrhage as well as chronic consequences of TBI [3]. CT is found to be the gold standard for the diagnosis of acute brain injuries because of its accessibility and speed, making it a popular first-line diagnostic modality. Detection and segmentation of traumatic lesions are crucial for appropriate medical decision-making [4]. Manual segmentation of TBI lesions on neuroimages by a radiologist can be time-consuming, expensive, and might be error-prone, particularly when a substantial volume of images needs to be evaluated [5]. As a result, there is a high demand for automatic computer-aided brain lesion segmentation and detection tools. However, neuroimages such as CT and MRI consist of non-brain tissue structures, including the dura, skull, eyeball,

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nose, and muscle. The existence of non-brain tissues poses a significant challenge for automated brain image processing. Non-brain tissue removal is a standard preprocessing step for most neuroimaging studies, enabling the proper identification of numerous brain-related conditions such as TBI, tissue classification, Alzheimer's disease, schizophrenia, and so on [6].

Skull stripping removes non-brain tissues, which is important for brain CT images. It improves accuracy in the analysis of subsequent studies like segmentation, feature extraction, and classification [7]. However, skull stripping is complicated for TBI images, brain segmentation can be difficult. One of them is the presence of traumatic lesions, which include hemorrhage and contusions that can alter the intensity distribution of brain regions, making the brain and non-brain distinction a challenge [3]. Furthermore, anatomical structures like optic nerves, eyes, and adjacent soft tissue have intensity values similar to brain structures, which makes segmentation challenging [8].

Traditional skull stripping methods, particularly thresholding-based approaches, often suffer from sensitivity to intensity variations and tend to either overstrip or leave non-brain tissues, especially in CT images with fractures or artifacts [3]. On the other hand, deep learning-based techniques, while powerful, require extensive annotated datasets and often fail to generalize across diverse scanner settings or clinical conditions like TBI, where deformities are common [7]. These limitations pose practical challenges in routine clinical settings. Our proposed method addresses these issues by using a contour-based and morphological approach that avoids the need for training data, adapts well to fractured skulls, and preserves the integrity of brain tissue for further analysis.

To address these complex challenges, this study proposes a novel skull-stripping method, ThresContCT\_TBI, specifically designed for use with TBI CT scans. As illustrated in Fig. 1, ThresContCT\_TBI integrates several well-established image processing strategies including intensity-based

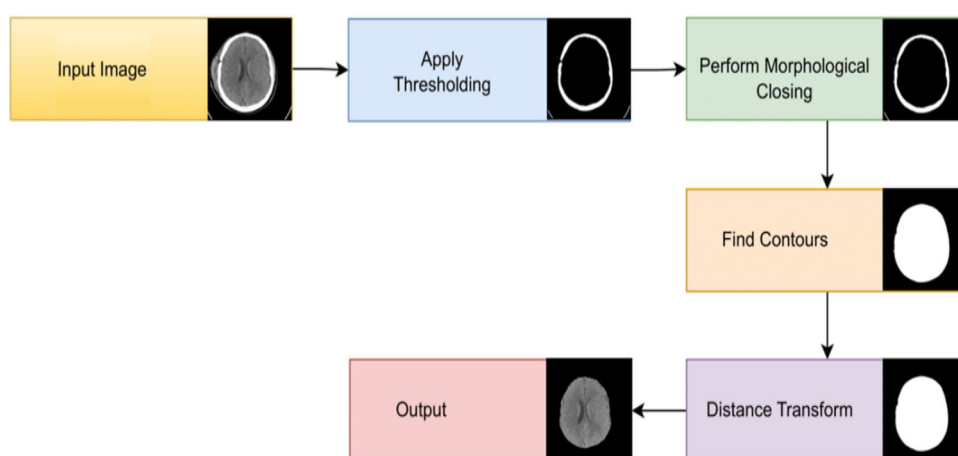
thresholding, morphological operations, and contour-based filtering with distance transform techniques to achieve robust and precise brain extraction. The method is engineered to be computationally efficient and adaptable to varying image quality levels, while effectively handling the heterogeneous nature of TBI-related structural abnormalities. The final skull-stripped results, as demonstrated in Fig. 2, show the method's capability to preserve essential brain regions while eliminating irrelevant non-brain tissues. This study is, to our knowledge, the first to introduce a traditional image processing pipeline tailored for skull stripping in TBI-specific CT datasets, offering a practical alternative where deep learning approaches struggle due to limited annotated data.

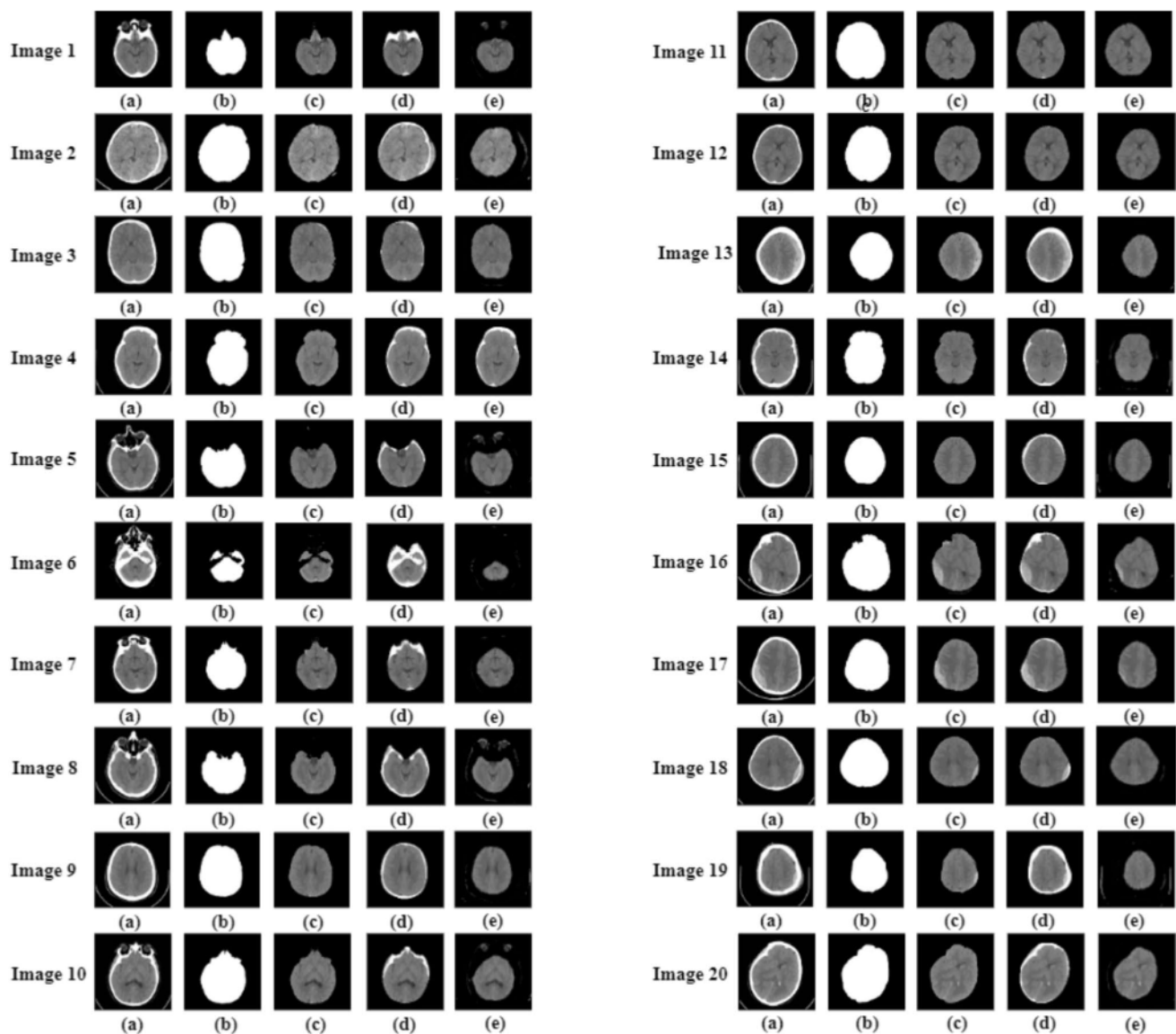
The main contributions of this study are as follows:

- We propose a lightweight, non-learning-based skull stripping approach, ThresContCT\_TBI, specifically designed for CT scans of TBI patients, where deep learning models often struggle due to limited annotated datasets.
- The method, ThresContCT\_TBI, combines adaptive thresholding, morphological processing, contour filtering, and distance-based refinement to effectively isolate brain tissue, achieving high segmentation accuracy without relying on training data.
- We provide quantitative results using standard metrics (accuracy, precision, recall, dice score, IoU, and HD) and comparisons with other existing methods to demonstrate the superiority and robustness of our approach.

The remainder of this paper is structured as follows: Section “[Related Work](#)” discusses a review of related previous work and existing skull-stripping techniques. Section “[Materials and Method](#)” elaborates on the materials and methods used in this study, including dataset details, preprocessing steps, and the proposed methodologies. The experimental framework described in Section “[Results and Discussion](#)” presents the experimental setup, performance evaluation

**Fig. 1** Skull Stripping Workflow: Steps for Isolating the Brain Region





**Fig. 2** (a) Original CT images, (b) ground truth brain masks, and (c) skull-stripped results using ThresContCT\_TBI, (d) skull-stripped results using U-Net, (e) skull-stripped results using DeepMedic

metrics, and a detailed discussion of the obtained results. Lastly, Section “[Conclusion](#)” concludes the paper and outlines potential future research directions aimed at improving automated brain image analysis for TBI and other neurological disorders.

## Related Work

The field of skull stripping has improved a lot throughout the years. Transitioning from basic methods to deep learning (DL). Traditional methods used rules like thresholding and region growing. Roy and Maji [9] used adaptive thresholding methods. They added morphological operations to

improve it. Several studies used the application of skull stripping to enhance the accuracy of brain MRI images. Benson and Lajish [10] improved skull stripping for MR images. They used morphological image processing for skull stripping. Laha et al. [11] used adaptive thresholding methods and mathematical morphological techniques to remove non-brain structures from MRI neuroimaging dataset. The method first smoothens the MRI image to reduce noise and enhance segmentation accuracy. Then, they applied an adaptive thresholding method that separates non-brain structures from the neuroimages. Morphological processing is used for the improvement of extracted images by filling the holes in the brain region. This approach is simpler and faster and shows strong performance on noisy MRI images. Rahmad,

F.R. et al. [12] worked on CT images. Their method combined intensity and morphology-based methods. These approaches achieve good results in terms of time and visualization. F. Segonne et al. [13] developed a hybrid technique for performing skull stripping in T1-weighted MRI scans. They combined the deformable surface models and watershed algorithms for automatic segmentation of brain tissues from non-brain structures.

Advanced strategies, such as machine learning approaches, have been utilized to address the drawbacks of traditional methods. Fatima et al. [14] provide an extensive review of automated skull-stripping approaches applied to brain MRI scans, comparing traditional methods to machine- and DL-based approaches. The paper shows that DL outperforms traditional and machine-learning approaches in several ways, such as accuracy and robustness. Using edge detection and filtering processes, Swiebocka-Wiek [15] introduced a morphology-based method to enhance brain tissue segmentation. The five-step methodology combines mixed thresholding, morphological processing, and edge detection to create a binary mask that isolates brain tissue from surrounding structures. Despite successfully removing skull structures, the technique requires further refinement to properly remove cerebrospinal fluid and reduce edge-sharpening artifacts. Somasundaram and Shankar [16] proposed a skull stripping method for brain MRI scans. In the work, the authors have used a clustering method to find brain boundaries inside the skull and then used a 2D region-growing algorithm to integrate all the identified clusters and remove the skull area along with regions outside the skull. BET (Brain Extraction Tool) [17] is a tool frequently used to separate brain and non-brain tissues obtained from MRI. The BET tool utilizes a deformative model for the process. BET2 [18] is the enhanced version of BET, which generates a better result based on a pair of T1- and T2-weighted MRI scans.

Mandle et al. [19] and Kumar et al. [20] have demonstrated the importance of skull stripping in CT images, where accurate brain region extraction enhances lesion detection and segmentation. Kumar et al. [20] employed skull stripping as a preprocessing step in their entropy-based unsupervised brain hemorrhage segmentation technique to improve segmentation accuracy. Mandle et al. [19] demonstrated a segmentation method that segments out the brain region from CT images to detect lesion boundaries and intracranial structures. This study highlights the importance of skull stripping for feature extraction in CT-based neuroimages for precise diagnosis.

Another major advancement is the application of pre-trained segmentation models for automated skull stripping. Nayyef and Al-Tammi [21] used Contrast Limited Adaptive Histogram Equalization (CLAHE) as a pre-processing

method to enhance the segmentation of brain tissues in MRI images to explore the use of DL methods. Kleesiek et al. [22] proposed a 3D CNN-based architecture for skull stripping on brain MRI scans containing brain tumors. The proposed method is promising for clinical use and large-scale studies, as results show high accuracy and faster processing speed while using a GPU. In another study Yogananda et al. [23] present a fully automatic deep learning based pipeline to segment the brain region from large-scale neuroimages. The pipeline starts with the pre-processing steps like data augmentation, resampling, and intensity normalization. The method used 3D U-net-based CNN model to remove non-brain structures from the MRI images and accurately segment the brain structures. To address limitations of manual and semi-automatic skull stripping methods, Tabassum et al. [24] proposed a novel deep learning method based on nnUNet for skull stripping in brain MRI. To evaluate the method, two publicly available datasets are used and the proposed method achieved an average Dice coefficient of 0.9960, a sensitivity of 0.9999, and a specificity of 0.9996. Most existing methods focus on MRI or rely on data-heavy DL approaches. Limited methods are optimized for CT in TBI with limited annotations, which our method addresses.

While DL and ML algorithms provide good results, but applications in CT images is complicated. As these models require a huge amount of labeled images. Getting labeled images is often scarce due to privacy regulations and inconsistencies in scanning protocols. Moreover, the computational demand of these models makes them less feasible for real-time clinical applications, particularly in resource-constrained settings. Unlike BET, BSE, or hybrid methods that depend on deformable models or learning-based estimators, our method remains fully rule-based, but leverages distance transforms and morphological steps in a carefully tuned sequence that balances precision and generalizability across CT scans with diverse lesions. Traditional techniques, such as morphology-based approaches and intensity thresholding, continue to hold relevance due to their efficiency, ease of implementation, and independence from large training datasets. Studies indicate that well-optimized traditional methods can achieve comparable results to advanced models in CT applications while ensuring robustness across varying imaging conditions. This is the first report, to our knowledge, of such a rule-based pipeline validated on a TBI dataset, demonstrating robustness where traditional MRI-oriented tools fail due to format (NIfTI) requirements or poor CT generalization. Thus, traditional approaches remain a valuable solution in clinical environments where computational resources are limited, and quick, reliable skull stripping is essential for diagnostic workflows.

## Materials and Method

### Dataset

We have used the “Computed Tomography Images for Intracranial Hemorrhage Detection and Segmentation” dataset, which is publicly available on PhysioNet [25]. The collection contains 82 CT images with a wide range of TBI conditions, such as cerebral contusions, subdural and epidural hemorrhages, skull fractures, and diffuse axonal injuries. The cases span varying levels of severity, from mild to severe, across a diverse patient population, with a mean patient age of 27.8 years, including both male and female subjects. Thus, the dataset inherently covers diverse TBI severities, lesion types, and imaging conditions, supporting the robustness of our method. All of the imaging data was obtained from the Al Hilla Teaching Hospital in Iraq, and it was scanned using a Siemens/SOMATOM Definition AS CT scanner with a slice thickness of 5 millimeters. Each image has a resolution of  $512 \times 512$  pixels with 16-bit gray-scale values. All images are in JPEG format.

The ground truth annotations for skull stripping are manually created and validated by two expert radiologists, one with 20 years and another with over 5 years of expertise in the field of neuroimaging. The annotations involved segmenting the brain region while excluding non-brain structures like skull, dura, and soft tissues.

### Proposed Skull Stripping Method

In this study, we introduce our skull stripping method (ThresContCT\_TBI), which is crucial as a preprocessing step in the analysis of brain CT images. ThresContCT\_TBI method employs Thresholding, morphological operations, Contour-based brain region mapping on CT images of TBI, and finally, border corrections using a distance transform to obtain the final brain region. This combination was chosen based on empirical evaluations across 40+ manually validated CT scans. We observed that combining distance transforms with morphological refinement consistently preserved brain regions while removing non-brain tissues, even in complex TBI cases. These operations are discussed next.

#### Thresholding for Skull and High-Intensity Structure Removal

The first step in skull stripping is to remove high-intensity structures, primarily the skull and dura mater, which appear brighter than the brain tissue in CT images. A range of threshold values  $\alpha$  is tested, and  $\alpha = 220$  is found to provide optimal separation between skull and brain region. Binary thresholding operation is applied with a threshold

value of 220, chosen empirically based on the intensity distribution in CT images. The definition of the thresholding function is given by Eq. 1:

$$T(x, y) = \begin{cases} 255 & \text{if } I(x, y) > \alpha \\ 0 & \text{if } I(x, y) < \alpha \end{cases} \quad (1)$$

Where  $I(x, y)$  denotes the intensity value of the pixel located at coordinates  $(x, y)$ . This step resulted in a binary mask in which the skull, dura, and other high-intensity structures were assigned a value of 255 (white), whereas all other pixels of the image were assigned a value of 0 (black) [26].

#### Morphological Processing for Mask Refinement

Although thresholding effectively removes the skull and dura, it leaves small gaps and discontinuities in the intracranial region. To address this, a morphological closing operation is performed using a  $20 \times 20$  kernel Eq. 2:

$$K = I_{20 \times 20} \quad (2)$$

Morphological closing (dilation followed by erosion) helps connect disjointed brain regions and fill small holes within the mask, ensuring that the extracted brain region remains intact [27].

#### Contour Detection and Non-Brain Region Elimination

After morphological refinement, the contours of the remaining structures are extracted using OpenCV's findContours function. However, the thresholded image still contained non-brain regions, such as parts of the eyes and extracranial soft tissues. To filter out irrelevant regions, a minimum area threshold of 500 pixels is applied [28].

$$A = \int_{\Omega} 1 \, d\Omega \quad (3)$$

Where in Eq. 3,  $A$  represents the area enclosed by a contour and  $\Omega$  denotes the region. Only contours with  $A \geq 500$  pixels are retained, ensuring that small noise artifacts are removed while preserving the brain.

#### Distance Transform-Based Refinement

To further refine the mask, a distance transform is utilized to enhance the separation between the brain and any remaining non-brain structures. The Euclidean distance transform computes the minimum distance from each foreground pixel to the corresponding nearest background pixel:



$$D(x, y) = \min_{(x', y') \in \text{background}} \sqrt{(x - x')^2 + (y - y')^2} \quad (4)$$

Here Eq. 4,  $D(x, y)$  refers to the distance value at pixel  $(x, y)$ . The function finds the nearest background pixel  $(x', y')$  and computes the Euclidean distance [29]. Pixels with lower  $D(x, y)$  values are closer to the background and likely belong to thin structures or the skull. A secondary threshold  $\beta$  is used to retain only deeper brain regions and remove unwanted outer components.

### Final Skull-Stripped Brain Extraction

The refined binary mask is then applied to the original CT image, which effectively removes the skull, dura mater, and eyes while preserving only the brain structure. A final filtering step is performed to eliminate high-intensity noise, where pixels with an intensity above  $\gamma$  were set to zero to ensure that any remaining non-brain artifacts were excluded.

The skull-stripped brain images were saved in a designated output directory for further analysis. Each processed image was also visualized for verification using Matplotlib (Fig. 2). The automated pipeline ensured consistent and reproducible skull-stripping across all CT images, facilitating subsequent brain-region-focused analysis.

## Results and Discussion

### Performance Evaluation Metrics

To assess the quantitative evaluation of the skull-stripping algorithm, we employed several segmentation evaluation indices, which include accuracy, precision, recall, DCS, IoU, and HD [30]. These metrics quantify the degree of overlap between the predicted and ground truth segmentation.

1. Accuracy: Accuracy measures the proportion of correctly classified pixels and is given by Eq. 5:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

where TP (true positives) are correctly segmented brain pixels, TN (true negatives) are precisely classified non-brain pixels, FP (false positives) represent misclassified non-brain regions, and FN (false negatives) correspond to brain pixels that were incorrectly removed.

2. Precision: Precision is defined as the ratio of correctly predicted brain pixels to all predicted brain pixels Eq. 6:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (6)$$

A higher precision value signifies a reduction in the false positive rate.

3. Recall: Recall, often referred to as sensitivity, measures the proportion of actual brain pixels correctly identified Eq. 7:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

A lower recall value suggests under-segmentation, where some brain regions are omitted.

4. Dice Coefficient (DSC): DSC, a widely used metric in the process of segmenting medical images, quantifies the overlap between the actual and predicted brain regions Eq. 8:

$$\text{Dice} = \frac{2TP}{2TP + FP + FN} \quad (8)$$

A Dice score close to 1 indicates a strong match between the ground truth and segmentation.

5. Intersection over Union (IoU): IoU, commonly referred to as the Jaccard Index, is calculated as Eq. 9. Higher IoU values indicate stronger overlap and hence better segmentation performance.

$$\text{IoU} = \frac{TP}{TP + FP + FN} \quad (9)$$

This metric measures the fraction of overlap between the segmented region and the ground truth.

6. Hausdorff Distance (HD): HD quantifies the worst-case deviation between the boundaries of the segmented mask and the ground truth. It is defined as:

$$HD(A, B) = \max \left\{ \sup_{a \in A} \inf_{b \in B} D(a, b), \sup_{b \in B} \inf_{a \in A} D(a, b) \right\} \quad (10)$$

$D(a, b)$  represents the Euclidean distance between points  $a$  and  $b$  from sets  $A$  (ground truth boundary) and  $B$  (segmentation boundary). A lower HD value indicates better boundary alignment.

## Results

The ThresContCT\_TBI method has been experimentally evaluated on the PysioNet dataset and validated using 6 indices. The computational experiments were conducted on a workstation equipped with a 12th-generation Intel core<sup>TM</sup>

**Table 1** Comparative Analysis of Skull Stripping Methods Based on Processing Time, Memory Usage, Dice Score, and IoU

| Method          | Time (s) | Memory Usage | Dice Score (%) | IoU (%) |
|-----------------|----------|--------------|----------------|---------|
| U-Net           | 5.54     | 4 GB(GPU)    | 82.4           | 70.8    |
| DeepMedic       | 7.96     | 5 GB(GPU)    | 66.6           | 50.6    |
| SegNet          | 5.18     | 3 GB(GPU)    | 57.03          | 40.43   |
| GrabCut         | 2.72     | 512 MB(CPU)  | 85.22          | 74.84   |
| ThresContCT_TBI | 0.62     | 280 MB(CPU)  | 97.79          | 95.68   |

i7-12650H 2.70 GHz, 16GB RAM, and an NVIDIA RTX 3080 GPU with 10 GB of VRAM. All image processing tasks, implemented in Python 3.9 using OpenCV 4.5.5, NumPy 1.22, and Matplotlib 3.5.1 libraries. Figure 2 depicts a subset of the dataset under study, wherein 20 images, their corresponding ground truth mask and their skull stripped output are reported. Our method takes 3 input parameters:  $\alpha$ ,  $\beta$ , and  $\gamma$ . From our exhaustive experiments, we report the range of these parameters as  $\alpha$  [200–240], and the threshold value  $\alpha$  is set to 220 for the removal of high-intensity structures. Similarly,  $\beta$  is set to 10 pixels, and  $\gamma$  is set to 240 to remove remaining non-brain artifacts. Table 2 summarizes the performance of the skull-stripping algorithm based on the above evaluation metrics, viz, accuracy, precision, recall, dice score, IoU, and HD. On average, our method processed each image in 0.62 seconds with peak memory usage of 280MB on a CPU setup, as summarized in Table 1. These results support its real-time feasibility in clinical environments.

The high accuracy of 98.68% confirms that the algorithm effectively differentiates between brain and non-brain regions. The Dice coefficient (97.79%) and IoU (95.68%) further demonstrate strong agreement between the segmented output and the ground truth. We observed precision of 98.43% indicates that the algorithm has a low false positive rate, meaning non-brain structures were effectively removed. However, the recall (97.16%) is slightly lower, implying that a small portion of the brain tissue might have been incorrectly excluded. This suggests a minor under-segmentation effect, possibly due to the thresholding step in the segmentation pipeline. The Dice coefficient (97.79%) and IoU (95.68%) confirm substantial overlap between the predicted segmentation and the ground truth masks. The Hausdorff Distance (60.86 pixels) is relatively high compared to

other metrics, indicating that some misalignment occurs at the boundaries of the segmented regions. This may be attributed to noise in intensity levels or variations in skull thickness across different images. Future improvements, such as edge-based refinement or active contour models, could help minimize boundary deviations.

A comparative evaluation was conducted between the ThresContCT\_TBI method and four widely used segmentation approaches, U-Net [31], the standard deep learning framework for biomedical image segmentation. DeepMedic [32], another widely used DL architecture for brain imaging tasks. SegNet [33], a lightweight CNN designed for efficient pixel-wise segmentation, and GrabCut [34], a classical graph-based algorithm requiring no training data. These methods were chosen because they are representative of both modern deep learning and traditional segmentation approaches, are compatible with our JPEG CT dataset, and provide a fair benchmarking context for evaluating the robustness of our proposed (ThresContCT\_TBI) lightweight method. As shown in Table 2, ThresContCT\_TBI outperformed the learning-based method in terms of accuracy (98.68%), precision (98.43%), Dice coefficient (97.79%), and IoU (95.68%), without requiring any training data. While U-Net and GrabCut achieved perfect recall and low HD, they demonstrated lower precision, indicating over-segmentation. DeepMedic and SegNet underperformed across most metrics. These results support the robustness and practicality of our approach, particularly for real-world clinical environments where labeled datasets and high computational resources are often limited.

The results obtained in this study are comparable to those achieved by DL-based skull-stripping methods in other datasets, which typically report Dice scores in the range of 98–99% [23, 24, 35, 36]. However, unlike DL models, ThresContCT\_TBI is not computationally intensive and does not require extensive training data, making it efficient and applicable in low-resource settings.

## Conclusion

This study presents a computationally efficient skull-stripping method, ThresContCT\_TBI, based on intensity thresholding, morphological operations, and distance

**Table 2** Comparative Evaluation of Skull Stripping Methods: Deep Learning Models, Classical Segmentation (GrabCut), and ThresContCT\_TBI

| Metric             | U-Net    | DeepMedic | SegNet   | GrabCut  | ThresContCT_TBI |
|--------------------|----------|-----------|----------|----------|-----------------|
| Accuracy           | 88.5%    | 78.1%     | 60.0%    | 90.63%   | 98.68%          |
| Precision          | 70.8%    | 58.5%     | 42.0%    | 74.84%   | 98.43%          |
| Recall             | 100%     | 78.3%     | 93.69%   | 100%     | 97.16%          |
| Dice Score         | 82.4%    | 66.6%     | 57.03%   | 85.22%   | 97.79%          |
| IoU                | 70.8%    | 50.6%     | 40.43%   | 74.84%   | 95.68%          |
| Hausdorff Distance | 15.96 px | 37.26 px  | 54.38 px | 14.11 px | 60.87 px        |

transformation techniques. The approach successfully isolates brain regions from CT images while preserving anatomical structures, achieving high segmentation accuracy. Performance evaluation using accuracy, precision, recall, Dice coefficient, IoU, and HD demonstrated robust segmentation results, with 98.68% of accuracy, a DSC of 97.79%, and an IoU of 95.68%. The results of these metrics demonstrate that there is a significant overlap between the predicted and ground-truth brain masks, which demonstrates that the method is successful in eliminating regions that are not part of the brain.

Despite these promising results, certain limitations exist. The recall value (97.16%), though high, suggests a slight tendency for under-segmentation, potentially excluding some brain regions. Additionally, the Hausdorff Distance (60.86 pixels) indicates minor boundary misalignments, which could be improved by refining segmentation edges. The ThresContCT\_TBI method is tested over CT images of TBI cases and normal brain CT images. However, its adaptability to other CT images, such as tumors and strokes, needs further testing with further tuning. Therefore, future work will extend validation to other neurological conditions. While deep learning models may better generalize across varied anatomical structures, our approach offers a lightweight and interpretable alternative where labeled data and computation are limited. This trade-off ensures practicality in low-resource environments. In future work, we aim to adapt this algorithm for direct DICOM image processing and integrate it into PACS-based (Picture Archiving and Communication System) clinical workflows, enabling seamless and fully automated skull stripping within hospital imaging systems.

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**Author Contributions** Minhazur Rahman performed the experiments, analyzed the data, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft. Bikash Upadhyay validated the data and approved the final draft. Rosy Sarmah conceived and designed the experiments, performed the experiments, analyzed the data, authored or reviewed drafts of the article, and approved the final draft.

**Data Availability** No datasets were generated or analysed during the current study.

## Declarations

**Competing interests** The authors declare no competing interests.

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